Efficient Fine-Tuning of Large Language Models

Abstract. This paper studies methods to reduce the cost of adapting large language models. We evaluate parameter-efficient fine-tuning (PEFT) techniques including LoRA, adapters and prefix tuning, together with 4-bit quantisation (QLoRA). Our objective is to quantify the trade-off between GPU memory, training time and downstream accuracy.

1. Methodology. We fine-tune a 7-billion-parameter model on three benchmarks: summarisation, question answering and sentiment classification. LoRA injects trainable low-rank matrices of rank 8 into attention projections while the base weights remain frozen. QLoRA additionally stores the frozen weights in 4-bit NormalFloat format. All experiments run on a single 24 GB GPU for three epochs.

2. Results. LoRA reduced the number of trainable parameters by 99.6 percent and cut GPU memory use by 62 percent relative to full fine-tuning. QLoRA reduced memory by a further 41 percent with an accuracy drop below one point on question answering. Training time per epoch fell from 9.1 hours to 3.4 hours. Sentiment classification accuracy remained within 0.5 points of full fine-tuning.

3. Limitations. Our study covers only English datasets and a single model family. We did not evaluate multilingual transfer, long-context tasks or the robustness of adapted models to adversarial prompts. Energy consumption was estimated rather than measured.

4. Future work. We plan to evaluate mixture-of-experts adapters, automated rank selection, and the combination of pruning with quantisation. Extending the evaluation to low-resource languages is an important direction.